

DOHEE KIM

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EXPERIENCE

- Software Developer (Co-op & Contract)
University Health Network (UHN)

July 2025 – Dec. 2025
Toronto, ON

 - Architected a reusable Week-Day-Level-Step content framework in Next.js/React/TypeScript, scaling delivery to 17 weeks (3 levels, 5 days/week) with consistent rendering and structure.
 - Developed one-click Excel export tool using Next.js, standardizing reporting columns for research stakeholders and reducing report generation time by 50%.
 - Automated user progression logic using Supabase (PostgreSQL) and Next.js API routes, replacing manual performance checks and saving more than 70 hours per week.
 - Implemented Jest unit testing within GitHub Actions pipelines, enforcing quality standards that blocked failed merges and reduced regressions.
- Software Developer Co-op
Korah

Sep. 2024 – Dec. 2024
Markham, ON

 - Engineered predictive models using Python State Machine algorithms to simulate 716 historical Neonatal Intensive Care Unit transfers, creating a standardized logic flow for bed allocation.
 - Visualized probability distributions and occupancy trends using Matplotlib, translating complex datasets into actionable research insights.
 - Leveraged Python (Pandas) to validate algorithm performance against manual outcomes, identifying a 50% optimization opportunity to reduce transfer distances and balance unit occupancy.
 - Collaborated with healthcare professionals to translate complex criteria into structured thresholds, reducing subjectivity and improving consistency in patient allocation.
- Software Developer (Co-op & Contract)
Locomobi

Jan. 2024 – May 2024
Mississauga, ON

 - Engineered a smart parking system using Python and machine learning based object detection (YOLO) to track vehicle occupancy in a 20+ capacity parking lot.
 - Structured SQL database schemas to optimize data pipelines for vehicle details, enabling accurate tracking of parking durations and enhancing system scalability.
 - Implemented fuzzy logic algorithms to resolve license plate recognition mismatches by incorporating entry/exit data, improving record accuracy to 95%.
 - Streamlined Agile workflows using Azure DevOps for task tracking and Git, implementing a CI pipeline for automated code reviews.

EDUCATION

- Sheridan College

Jan. 2023 – Dec. 2025

Advanced Diploma in Software Development and Network Engineering

Oakville, ON

TECHNICAL SKILLS

Languages: Python, Java, C++, SQL, JavaScript, TypeScript, PHP
Frameworks/Tools: Spring Boot, React, Flask, Angular, Bootstrap, Git, Linux, MQTT
Databases/Cloud: MySQL, PostgreSQL, MariaDB, MongoDB, AWS, Azure

PROJECTS

- Smart Waste Management System | *Angular, Spring Boot, Java, MQTT, MariaDB, Raspberry Pi*

 - Awarded Top 10 of 49 teams in the 2025 Sheridan Capstone Showcase, recognized for project excellence, innovation, and technical skill.
 - Constructed an interactive Angular frontend to visualize sensor data and manage bin statuses through a responsive dashboard.
 - Built 8+ REST controller endpoints supporting CRUD operations, cleaner to bin assignments, and entity relationship handling to prevent data inconsistencies.
 - Configured an MQTT publish/subscribe workflow on a Raspberry Pi to reliably ingest ultrasonic sensor data in real time
- ReefRiches (ElleHacks) | *React.js, JavaScript, Google Gemini API, CSS, Git*

 - Developed a gamified financial literacy web app for children ages 7-12 using React.js, delivering a responsive, component-based UI for an interactive learning experience.
 - Integrated Google Gemini API and designed prompts to produce structured JSON responses, enabling reliable parsing and dynamic content rendering on the frontend.
 - Implemented React Context API and useReducer to handle asynchronous game logic, synchronizing real-time player progression and persistent module states.
 - Led AI feature implementation in a team of four, shipping a functional MVP within a 36 hour hackathon.